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Criminalistics

Validating strength-of-support conclusion scales for fingerprint, footwear, and toolmark impressions

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Abstract

In the pattern comparison disciplines such as fingerprints, footwear, and toolmarks, the results of a comparison are communicated by examiners in the form of categorical conclusions such as Identification or Exclusion. These statements have been criticized as requiring knowledge of prior probabilities by the examiners and being overinterpreted by laypersons. Alternative statements based on strength-of-support language have been proposed. The current study compares traditional conclusion scales against strength-of-support scales to determine how these new statements might be used by examiners in casework. Each participant completed 60 comparisons within their discipline, which were designed to approximate casework conditions, using either a traditional or a strength-of-support conclusion scale. The scale used on each trial was randomly assigned, and participants knew the scale for that trial as they began the comparison. Fingerprint examiners were much less likely to use Extremely Strong Support for Common Source than Identification. Footwear examiners treated the traditional and strength-of-support scales similarly, but toolmark examiners were much less likely to use Extremely Strong Support for Common Source than Identification, similar to fingerprint examiners. A separate group of fingerprint examiners used Identification less often when an expanded scale was available. The results demonstrate that expanded scales may result in the highest conclusion category being used less often by examiners when other alternatives are possible, and the term "extremely strong support" may introduce risk aversion on the part of examiners.

KEYWORDS
articulation language, categorical conclusions, fingerprints, footwear, strength of support, toolmarks

Highlights

- Fingerprint, footwear, and tool mark examiners were tested in simulated casework.
- Participants used either categorical conclusion or strength-of-support scales when reporting conclusions.
- Fingerprint and tool mark examiners used Extremely Strong Support for Common Source less often than Identification.
- Footwear examiners treated traditional and strength-of-support scales equivalently.

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- Fingerprint examiners redefined the Identification conclusion when the traditional scale was expanded.
- Strength-of-evidence scales may make examiners more risk-averse but may provide as investigative leads.
- Policy makers should conduct interal testing of proposed changes to articulation language prior to implementation.

1 | INTRODUCTION

Practitioners working in the pattern comparison disciplines in the US traditionally express their conclusions using an articulation statement drawn from a small set of categorical conclusions. Quantitative tools for assessing the strength of support exist in some disciplines, but the vast majority of pattern comparisons are performed by human experts, who conduct manual comparisons between two or more impressions to determine whether the two patterns might share a common origin. In friction ridge comparisons, the determination of the examiner is typically expressed using a categorical conclusion such as "Identification," "Inconclusive," and "Exclusion" and then provided to stakeholders such as police, lawyers, detectives, jurors, or judges. Footwear and tool mark comparison disciplines use similar conclusions scales but with additional categories, in part because the manufacturing process creates features that tend to be similar (i.e., "repeated" features as opposed to "unique" features that are created through use or wear).

Categorical conclusions have been criticized on several fronts, in part because they represent a statement about a posterior probability rather than a statement about the strength of the support for different propositions. They are posteriors because they are a statement about a proposition (e.g., Identification: the two impressions were made by the same finger) rather than a statement about the evidence *given* a proposition (e.g., the likelihood of observing this amount of detail in agreement given a same source proposition is high). The latter statement is a likelihood and requires no information about the prior probabilities of mated or nonmated pairs. However, the categorical conclusion does require consideration of the priors: in the extreme, if the examiner is never provided with a mated pair, the examiner should never reach an Identification determination regardless of the strength of the evidence. However, the likelihood statement does not suffer from this problem because the consumer will multiply the likelihood by the prior to produce their own posterior, and if they believe the prior probability of a mated pair is low or zero, they can simply adjust accordingly.

Jackson et al. [1] argued for strength-of-support approaches such as a likelihood ratio in all forensic disciplines because such an approach does not presume a prior probability and allows the trier of fact to update their beliefs in a transparent way. In addition, categorical statements such as Identification are prone to overinterpretation by laypersons. Swofford and Cino [2] assessed the beliefs of potential jurors and found that 71% of those surveyed interpreted "identification" to imply "to the exclusion of all others." Thus, there appears to be a disconnect between what examiners intend and how their conclusions are interpreted. In this case, jurors interpret the evidence as stronger as was originally intended by the examiner.

Although pattern disciplines are just beginning to add quantitative tools that might provide a numerical estimate of the strength of the evidence, policymakers have begun to explore strength-of-support type language to avoid posterior conclusions. The present work explores the consequences of adopting this new language.

There is a robust debate in the literature about the use of definitive conclusions such as Identification. For example, the International Association for Identification held a symposium at their annual meeting in Atlanta in 2017 that considered whether the term Identification should be used as a conclusion. While that symposium produced no consensus or strong momentum for change within the forensic community, others outside the community have called for a shift away from definitive statements such as Identification to statements that express the strength of the evidence such as likelihood ratios or verbal equivalents [3–5]. Some authors have argued that forensic examiners should not even make sole-source statements (e.g., [6], [7]) and instead argued for a strength-of-evidence framework for conclusions, with language such as "Supports" rather than definitive conclusions such as "Identified." However, whether jurors can understand more complex statistical terms such as likelihood ratios and random match probabilities is an empirical question with no clear answer in the literature [1]. In addition, jurors might benefit from having examiners make definitive statements because examiners may have a better understanding of the context of the examination. There have been cogent arguments on both sides, but the literature contains relatively few direct tests of how examiners would change their choice of conclusions should they be asked to adopt a new set of conclusion statements.

In the fingerprint discipline, one potential solution to the issues raised above is to keep the traditional conclusion language but expand the scale to include some statements that are phrased as strength of support. Prior work on this question by our group revealed a surprising result: the interpretation of the term Identification by fingerprint examiners depended on the scale in which it was embedded [8]. We presented 60 casework-like

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comparisons to 27 latent print examiners and asked them to use either the traditional 3-conclusion scale or an expanded one that included *support for common source* and *support for different sources*. Examiners knew on each trial which scale they would use, and we fit the data using an extension of signal detection theory [9], which provides separate estimates of examiner ability (through an estimate of signal detection's sensitivity measure) and examiner risk aversion (through an estimate of decision criteria). Response bias can be thought of as how risk-averse a participant is when making decision: an examiner who makes more Identification conclusions than their colleagues will make more correct decisions but also has an elevated risk erroneous identification outcomes [10]. Results from both the raw data and the signal detection modeling fits demonstrated that examiners used the Identification conclusion less often when the Support for Common Source conclusion was available. This means that examiners redefined what was sufficient for an Identification conclusion when they had the Support for Common Source conclusion available as part of the conclusion scale. They also redefined the Inconclusive term, using it less often when Support for Common Source and Support for Different Sources was available.

The present work extends this approach to comparisons between a traditional scale and a pure strength-of-support conclusion scale in fingerprint, footwear, and toolmark disciplines. Strength-of-support scales have the potential advantage that they focus on the evidence rather than the conclusion of an examiner. A traditional conclusion such as Identification has a definitive nature to the statement and is interpreted as such [2]. In fields such as DNA where likelihood ratios are common, the evidence is framed in terms of the probability of the observations *given* the hypothesis of same source and the probability of the observations *given* the hypothesis of different sources. This allows the jury to evaluate the evidence in conjunction with the rest of the case without the expert making the decision that is really in the domain of the jury. Strength-of-support statements also have the potential to seem less definitive, which may be appropriate given the low but nonzero erroneous identification error rates [11]. As a result, governing bodies such as the Organization of Scientific Area Committees for Forensic Science (OSAC) are considering new language that relies on strength-of-evidence statements such as Extremely Strong Support for Common Source [12]. The traditional scale for likelihood ratios typically maps this verbal statement to likelihood ratios of 1000–10,000 in the forensic disciplines [4], and this is consistent with black box studies that show error rates of around 0.1% for fingerprints [11].

The goal of the present work is to validate proposed strength-of-support scales, to measure how fingerprint, footwear, and tool mark examiners would use new scales in casework-like situations. An additional comparison with fingerprint examiners addresses whether strength-of-support language might be added to existing traditional scales. Together, the combined work allows for a complete calibration of the language such that it accurately reflects the intent of the examiner and is interpreted properly by the layperson.

2 | METHOD

Participating forensic examiners each conducted 60 casework-like comparisons in their discipline of expertise. On each trial, they were give either the traditional scale from their discipline or a scale based on strength-of-support language. Some fingerprint examiners participated in a conceptual replication of the Carter, Vogelsang [8] study. The present study was conducted using web-based interfaces written in Javascript, with data stored remotely in a MySQL server. All data were collected according to a Human Subject protocol approved by Indiana University.

2.1 | Participants

For the fingerprint portion of the study, 66 latent print examiners from forensic facilities participated. They were required to be eligible to testify in the United States. This portion of the study had two groups of participants. Thirty-two examiners compared the traditional scale with an expanded traditional scale, while 34 examiners compare the traditional scale with a pure strength-of-support scale. Of the participants who completed demographic surveys, 50 were female, 13 were male, and one declined to answer. The median age was 40, with an age range of 27 to 62. Twenty-one had no eye correction, 12 had contacts, 27 had glasses, and 4 had Lasik. Twelve worked in Federal agencies, 20 worked in local agencies, 10 worked in metro/county agencies, 17 worked in state agencies, 3 worked in other agency types, and 2 preferred not to answer.

For the footwear portion of the study, 32 footwear examiners from forensic facilities participated. They were required to be eligible to testify in the United States. Of the participants who completed demographic surveys, 25 were female and 6 were male. The median age was 43, with an age range of 27–71 years. Eight had no eye correction, 8 had contacts, 10 had glasses, and 5 had Lasik. One worked in a Federal agency, 5 worked in local agencies, 4 worked in metro/county agencies, 20 worked in state agencies, and 1 worked in other agency type.

For the toolmark portion of the study, 20 toolmark examiners from forensic facilities participated. This subject count is lower than fingerprint and footwear datasets due to the Covid-19 pandemic, but still sufficient for data analysis. Toolmark examiners were required to be eligible to testify in the United States. Of the participants who completed demographic surveys, 7 were female and 10 were male. The median age was 42, with an age range of 28–54 years. Five had no eye correction, 4 had contacts, 7 had glasses, and 2 had Lasik. Four worked in local agencies, 6 worked in metro/county agencies, and 8 worked in state agencies.

2.2 | Stimuli

All stimuli were collected under the supervision of a subject matter expert (Vanderkolk) with the goal of making the trials similar to

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casework. All images used in the study are available for inspection from the OSF site linked below.

2.2.1 | Fingerprints

Fingerprint impressions were selected from a 3000-print database collected from volunteering Indiana University staff and students. Each exemplar print was labeled with an anonymized participant code and the hand and finger the print was from, then scanned into an editing software. All exemplar prints were tapped or rolled ink prints. The latent prints were black powder developed on clear plastic tumblers and lifted with lift tape, ninhydrin developed on paper, black powder on galvanized metal, or ink prints on white paper. The latent prints were also labeled with a participant code and the hand and finger, then scanned into the same editing software to create the database. Images were scanned using an Epson Perfection 4870 scanner at 4800 pixels per inch and downsampled to 750 pixels per inch. The final display resolution was dependent on the size of the participant's monitor, but a higher resolution could be accessed through a zoom function.

The latent prints chosen for the study contained various sources of noise such as distortion, scarring, smearing, medium, contrast, and percentage of print present, while the exemplar prints were typically of high quality. Our goal was to create a test set of stimuli that were similar to other error rate studies (e.g., Ulery, Hicklin [11]), although we do not consider this study to measure error rates but instead provide a comparison of two reporting scales under conditions that are similar to casework. To that end, we selected our nonmated images using left–right reversed impressions from the opposite hand of the donor individual. We used a subject matter expert (Vanderkolk) to select both mated and nonmated pairs that were similar in difficulty to what examiners experienced during typical casework. Thus, our exemplar impressions for nonmated pairs were designed to be challenging exclusions that for the most part bore superficial similarity to the latent impression.

2.2.2 | Footwear

Footwear stimuli were collected under the guidance of our subject matter expert (Vanderkolk). We used a collection of shoes drawn from different sources. Half of the trials contained shoes that were the same make and model purchased by a runner who wore them to approximately the same wear level. The other half of the trials contained shoes and light hiking boots that were chosen because at least two pairs of the same make and model were available, and there were 9 different models, some with multiple exemplars. All shoes had been moderately worn. Similar soles with similar amounts of wear were used to produce challenging impressions for the study. Only heel impressions were used for the study because image acquisition proved to be easier to manage and more reliable. All images used in the study are available from the OSF site linked below.

Shoe prints, or impressions, were made using different techniques to produce images bearing various qualities and quantities of details. One technique consisted of applying extremely light to somewhat heavy mixtures of petroleum jelly and black fingerprint powder to the soles of the shoes. This mixture was applied to gloved fingers then gently rubbed onto the sole. The soles of the shoes were then pressed, rolled, or slapped onto pieces of white paper. The other technique consisted of applying melted chocolate ice cream to the soles of the shoes. Melted chocolate ice cream was chosen to produce a dark viscous matrix that dried quickly for the impressions. With the melted chocolate ice cream on a flat paper plate, the soles were tapped into the ice cream. The soles of the shoes were then pressed, rolled, or slapped onto pieces of white paper. Some of the impressions either had been made from areas of the soles that were relatively clean or areas that had been previously used to make the petroleum jelly/powder impressions. These techniques produced various qualities and quantities of recorded details in the impressions. Areas from the prints were selected to produce comparison pairs that ranged from easy to difficult.

Petroleum jelly and black powder impressed on a white piece of paper is a standard method for creating standard shoeprints for comparison, and this was done for each heal first. We then impressed the heal of each shoe into a matrix of nearly dried chocolate ice cream, which is a tacky, syrupy consistency, and makes a visual characteristic somewhat similar to blood. We chose this method because it is safer to handle than blood. If we had used animal or personal blood, the process would have been cumbersome to create near-threshold impressions and we would have had to follow extensive safety protocols within our institution. The goal of this study is to compare across scales, and so, if this matrix made the images easier or harder than casework, it would affect both scales equivalently. While this is a limitation of the study with respect to how similar the images might be to casework, we invite the reader to view the images in the GitHub site to determine for themselves whether the images are representative of what might be encountered in casework.

Once dried, the impressions were scanned at 600 pixels per inch using an Epson V600 scanner. Images were then downsampled to 200 pixels per inch. Photographs of the known shoe image were taken with a Sony α7IIIr camera with a FE 1.4/24 GM lens and downsampled to match the pixels per inch of the scanned images. The final display resolution was dependent on the size of the participant's monitor, but a higher resolution could be accessed through the zoom function.

Mated pairs were created by pairing the questioned images from the simulated crime scene methods with the gel lifts and photographs of the same shoe. Nonmated pairs were created by using impressions from the same make and model shoe yet different shoes. The chosen shoes were potentially quite difficult due to the fact that not only were they the same make and model, but some had been worn by the same individual and therefore likely subject to the similar, but still different, wear patterns. Because of this, we do not consider this dataset to accurately represent error rates in the field. Instead, the goal is to identify differences between the two scales under comparison, and so, we merely require a dataset that

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is somewhat similar to casework. All images used in the study are available from the OSF site linked below.

2.2.3 | Toolmarks

Striated toolmarks were collected from 15 quarter-inch screwdrivers (Craftsman 9–41,584 1/4" x 6" Slotted Screwdriver) and 15 quarter-inch wood chisels (TEKTON 67551 1/4-Inch Wood Chisel). These were purchased in the same order from Amazon, although we have no control over the batch origin. We constructed a custom 3D-printed jig that was used to produce striated scrapings in heavy-duty aluminum foil (see Figure 1), and while we collected scrapings at 5 different angles (10, 20, 35, 55, and 80 degrees), we judged the 20- and 35-degree angles to be most representative of what might be produced by tools used on metal window and door frames to gain access to property. Thus, we used only scrapings collected using either of these two angles. Each tool was used to create three separate scrapings at each angle, and care was taken to mark the tool face used to create the marks because flat screwdrivers are double-sided. Chisel blades were single-sided. The toolmarks were lit with oblique lighting using a fiber optic light source and photographed with a Sony α7IIIr camera and FE 2.8/90 Macro G OSS lens with two Kenko DG extension tubes totaling 26-mm extension to improve image capture size. The images were then downsampled to 1500 pixels per inch. The final display resolution was dependent on the size of the participant's monitor, but a higher resolution could be accessed through the zoom function.

Mated pairs were created by selecting one scraping from each tool at either 20 or 35 degrees and then presenting the 20 and 35 degree images from the same tool from a different scraping. Nonmated pairs were created by selecting similar-looking scrapings from different tools.

While it is difficult to determine whether the task difficulty was comparable to typical casework, the goal of the experiment is to compare two scales, and thus, we simply need the task difficulty to be generally similar to casework. All images used in the study are available from the OSF site linked below.



FIGURE 1 Custom 3D-printed jig for making toolmark impressions [Color figure can be viewed at wileyonlinelibrary.com]

2.3 | Instructions

The instructions for all three sets of participants included descriptions of the different scales. The definitions for each scale are found in Table 1 and Table 2 for fingerprint examiners, Table 3 for Footwear examiners, and Table 4 for Toolmark examiners. In addition to these definitions, the instructions included the following general statements:

2.3.1 | Fingerprints

Within the field of latent print identification, various groups, including the Friction Ridge Subcommittee of OSAC, are contemplating changes to the way that conclusions are reported. The groups are proposing additional categories beyond the traditional Identification/Inconclusive/Exclusion conclusions that have historically been used. The goal of this experiment is to understand the consequences of moving to different conclusions scales, and we are testing scales that have some language in common with the Draft Standard for Friction Ridge Examination Conclusions as produced by the Friction Ridge Subcommittee of OSAC.

2.3.2 | Footwear

Within the pattern comparison disciplines, various groups are contemplating changes to the way that conclusions are reported, including a shift to language that expresses conclusions according to the strength of support for one of two propositions (Common Source or Different Sources). The goal of this experiment is to understand the consequences of moving to conclusion scales that express conclusions according to strength of support for propositions. These strength-of-support statements are alternatives to definitive statements such as Identification or Exclusion. However, before we mandate new language, we need to understand the consequences of adopting new conclusion language and thus this experiment.

2.3.3 | Toolmarks

Within the field of firearm and toolmark comparisons, various groups, including the Firearms & Toolmarks Subcommittee of OSAC, are contemplating changes to the way that conclusions are reported. However, before any change is made, we need to understand the consequences of such a change. The goal of this experiment is to understand the consequences of moving to different conclusion scales.

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TABLE 1 Traditional and Expanded Traditional statements that friction ridge examiners were asked to use during casework-like comparisons

Traditional scale	Expanded traditional scale
Identification: Identification is the strongest degree of association between two friction ridge impressions. It is the conclusion that the observations provide extremely strong support for the proposition that the impressions originated from the same source and extremely weak support for the proposition that the impressions originated from different sources. Identification is reached when the friction ridge impressions have corresponding ridge detail and the examiner would not expect to see the same arrangement of details repeated in an impression that came from a different source.	**Identification**: Identification is the strongest degree of association between two friction ridge impressions. It is the conclusion that the observations provide extremely strong support for the proposition that the impressions originated from the same source and extremely weak support for the proposition that the impressions originated from different sources. Identification is reached when the friction ridge impressions have corresponding ridge detail, and the examiner would not expect to see the same arrangement of details repeated in an impression that came from a different source.
	Support for Common Source: Support for Same Source is the conclusion that the observations provide more support for the proposition that the impressions originated from the same source rather than different sources; however, there is insufficient support for an Identification.
Inconclusive: The observed characteristics of the items are insufficient to support any of the other conclusions (including one of the "support" conclusions if they are available).	**Inconclusive**: The observed characteristics of the items are insufficient to support any of the other conclusions (including one of the "support" conclusions if they are available).
	Support for Different Sources: Support for Different Sources is the conclusion that the observations provide more support for the proposition that the impressions originated from different sources rather than the same source; however, there is insufficient support for an Exclusion.
Exclusion: Exclusion is the conclusion that two friction ridge impressions did not originate from the same source. Exclusion is reached when in the examiner's opinion, considering the observed data, the probability that the two impressions came from the same source is considered negligible.	**Exclusion**: Exclusion is the conclusion that two friction ridge impressions did not originate from the same source. Exclusion is reached when in the examiner's opinion, considering the observed data, the probability that the two impressions came from the same source is considered negligible.

TABLE 2 Traditional and Strength of Support statements that friction ridge examiners were asked to use during casework-like comparisons. Definitions were not provided for the traditional scale for this group of participants, but instead they read the following instructions: "For the traditional categories of Exclusion, Inconclusive, and Identification, we would like you to use the criteria that you use in casework. You may choose from Exclusion, Inconclusive, or Identification on each trial for your conclusion"

Traditional	Strength of support
Identification	**Extremely Strong Support for Common Source:** Extremely Strong Support for Common Source is the strongest degree of association between two friction ridge impressions. It is the conclusion that the observations provide extremely strong support for the proposition that the impressions originated from the same source and weak or no support for the proposition that the impressions originated from different sources. This conclusion is reached when the friction ridge impressions have corresponding ridge detail, and the examiner would not expect to see the same arrangement of details repeated in an impression that came from a different source.
	Support for Common Source: Support for Common Source is the conclusion that the observations provide more support for the proposition that the impressions originated from the same source rather than different sources.
Inconclusive	**Inconclusive**: The observed characteristics of the items are insufficient to support any of the other conclusions.
	Support for Different Sources: Support for Different Sources is the conclusion that the observations provide more support for the proposition that the impressions originated from different sources rather than the same source.
Exclusion	**Extremely Strong Support for Different Sources**: Extremely Strong Support for Different Sources is the conclusion that the observations provide much more support for the proposition that the impressions originated from different sources and weak or no support for the proposition that the two items originated from the same source.

2.4 | Procedure

2.4.1 | Fingerprints

The study was composed of 60 trials for each participant, and each trial consisted of one fingerprint comparison. The experiment was administered electronically using a custom Javascript interface designed to mimic the tools available during casework (see Figure 2). On each trial, the latent print was placed on the left side of the screen next to an exemplar print on the right side, as shown in Figure 2. The interface allowed the participants to zoom, rotate, and pan the individual images, as well as mark individual features with transparent digital markers.

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TABLE 3 Definitive conclusion and strength-of-support statements used by footwear examiners

Definitive conclusions	Strength of support
Identification: The footwear impressions correspond in physical size, design, class, wear, and randomly acquired characteristics. The likelihood of observing this quality and quantity of correspondence if the questioned impression was made by a different source is considered extremely low.	**Extremely Strong Support for Common Source**: The questioned impression and the impression from the known footwear share sufficient quality and quantity of agreement of class, wear, and randomly acquired characteristics. The observed characteristics provide extremely strong support for the proposition that the questioned impression was made by the known footwear **and** little to no support for the proposition that the questioned impression was made by a different source.
High Degree of Association: The footwear impressions appear to have strong associations; however, the quality and quantity of shared characteristics are insufficient for an identification. Other footwear with the same class characteristics as observed in the known impression are included in the population of possible sources only if they display similar wear and randomly acquired characteristics as observed in the questioned impression.	**Strong Support for Common Source**: The observed characteristics exhibit strong associations between the questioned impression and the impression from the known footwear. These characteristics offer stronger support for the proposition that the questioned impression came from the known footwear than for the proposition that the questioned impression came from another source. Other footwear with the same class characteristics as observed in the known impression are included in the population of possible sources only if they display similar wear and randomly acquired characteristics observed in the questioned impression.
Limited Association: The footwear impressions correspond in size and shape of class characteristics. Other footwear having similar class characteristics may be included as possible sources.	**Support for Common Source**: The questioned impression and the impression from the known footwear correspond in class characteristics. The observed characteristics of the items provide more support for the proposition that the questioned impression came from the known footwear than for the proposition that the questioned impression came from another source. Other footwear with the same class characteristics as observed in the known impression are included in the population of possible sources.
Inconclusive: Evaluation of the footwear impressions is inconclusive due to insufficient data to support an inclusion or exclusion conclusion of the shoe as a possible source.	**Indeterminate With Respect to Source**: The observed characteristics are insufficient or too ambiguous to support any other source conclusions, as defined in the other sections, or support the two competing propositions equally.
Indications of Non-Association: The footwear impressions have dissimilarities which indicate non-association; however, the details or features are not sufficient to permit an exclusion.	**Support for Different Sources**: The questioned impression exhibits dissimilarities when compared to the known footwear and provide stronger support for the proposition that the questioned impression came from a different source than the proposition that the questioned impression came from the known footwear.
Exclusion: The two impressions originated from different footwear.	**Extremely Strong Support for Different Sources**: Sufficiently significant differences were noted in class tread design or sufficiently significant differences were noted in the comparison of wear or randomly acquired characteristics between the questioned impression and the impression from known footwear to state that the known footwear is not capable of having made the questioned impression. (Such as, there is significantly different wear or randomly acquired characteristics between the impressions, especially when there is more wear or randomly acquired characteristics in the questioned impression than the known impression.)

Each trial began with an "of value" decision, which in casework allows the examiner to decide not to proceed with a comparison due to poor quality of the latent impression. However, while we recorded this response, we still required the participant to complete the trial. We made this decision because the interpretation of our results depends in part on model fits from signal detection theory, and it is difficult to fit models in which an initial quality threshold is assessed. Both scales included an "inconclusive" category, and while we understand that in casework "no value" and "inconclusive" have different meanings, we considered the two to be approximately equal for the purposes of comparing two conclusion scales. However, we also report data where we exclude the trials that were judged to be of no value. Note that our goal is to compare across different conclusion scales, not to measure error rates, and given that the "no value" decision was made prior to knowing which scale would be used on any particular trial, the differences in use of the "no value" decision cannot meaningfully differ across the two scales except for statistical noise.

We randomized the assignment of images to conditions (3-conclusion and 5-conclusion scales) across participants, and thus, we would not expect a systematic bias of image quality on one of the two scales. This is an important consideration because if we find that one scale produces more risk-averse use of the highest category of responses, this cannot be due to the fact that scale was assigned more difficult images.

After making an "of value" determination, the exemplar impression also became visible. Data collection was terminated after

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TABLE 4 Definitive conclusions and strength-of-support statements used by toolmark examiners

Definitive conclusions	Strength of support
Identification: Agreement of all discernible class characteristics and sufficient agreement of a combination of individual characteristics where the extent of agreement exceeds that which can occur in the comparison of toolmarks made by different tools and is consistent with the agreement demonstrated by toolmarks known to have been produced by the same tool.	**Extremely Strong Support for Common Source:** Extremely Strong Support for Common Source is the strongest degree of association between two tool marks. It is the conclusion that the observations provide extremely strong support for the proposition that the tool marks originated from the same source and weak or no support for the proposition that the tool marks originated from different sources. This conclusion is reached when the tool marks have corresponding detail and the examiner would not expect to see the same arrangement of details repeated in a tool mark that came from a different source. This conclusion implies agreement of all discernible class characteristics and therefore the basis for this conclusion comes from the observed individual characteristics.
Insufficient for Identification: Agreement of all discernible class characteristics and some agreement of individual characteristics, but insufficient for an identification.	**Support for Common Source:** Support for Common Source is the conclusion that the observations provide more support for the proposition that the tool marks originated from the same source rather than different sources. This conclusion implies agreement of all discernible class characteristics and therefore the basis for this conclusion comes from the observed individual characteristics.
Inconclusive: Agreement of all discernible class characteristics without agreement or disagreement of individual characteristics due to an absence, insufficiency, or lack of reproducibility.	**Inconclusive:** The observed characteristics of the items are insufficient to support any of the other conclusions. This conclusion implies agreement of all discernible class characteristics and therefore the basis for this conclusion comes from the observed individual characteristics.
Insufficient for Elimination: Agreement of all discernible class characteristics and disagreement of individual characteristics, but insufficient for an elimination.	**Support for Different Sources**: Support for Different Sources is the conclusion that the observations provide more support for the proposition that the tool marks originated from different sources rather than the same source. This conclusion implies agreement of all discernible class characteristics and therefore the basis for this conclusion comes from the observed individual characteristics.
Elimination: Significant disagreement of discernible class characteristics and/or individual characteristics.	**Extremely Strong Support for Different Sources:** Extremely Strong Support for Different Sources is the conclusion that the observations provide much more support for the proposition that the tool marks originated from different sources and weak or no support for the proposition that the tool marks originated from the same source. This conclusion can be made on the basis of either class characteristics or individual characteristics.



FIGURE 2 Interface used by fingerprint examiners to conduct casework-like comparisons [Color figure can be viewed at wileyonlinelibrary.com]

30 min to avoid participant dropout because 60 trials is a lot to ask of participants. Note that our primary focus is on a comparison of the two scales assigned to each participant, rather than measuring errors or determining an error rate. This makes it less critical that testing conditions exactly mirror casework. We also included a "Pause" button available that hid the trial and paused the countdown timer until the "Resume" button was selected. Participants indicated their decision using buttons at the bottom of the screen. If the timer ended before they made a decision, then the prints were hidden, and they were asked to make a decision before continuing to the next trial. Participants completed 30 trials using the 3-conclusion scale and 30 trials using the 5-conclusion

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scale. Images were randomly assigned to condition for each participant, and the order of the images and conditions was randomized for each participant. As a result of this randomization, we would not expect our results to be affected by, say, difficult trials only being assigned to the traditional scale. Half of the trials were designated as mated pairs, and half were nonmated.

Participants received only the instructions and training provided by the text in either Table 1 or Table 2 depending on the comparison they were randomly assigned and did not have extensive training on the new categories in the expanded or strength-of-support scales. We acknowledge that the behavior of examiners may change as they adapt to the use of novel statements if they were to be included in operational casework. For the participants in the comparison between Traditional and Expanded Traditional Scales, both scales use the "Identification" and "Exclusion" categories as shown in Table 1, which examiners have had experienced with and presumably should not change, although this is an empirical question. Participants in the traditional and strength-of-support comparison condition were provided the definitions of the terms shown in Table 2.

There was one other difference between the two Fingerprint participant groups: The comparisons for the two groups are asking slightly different scientific questions. The first comparison conceptually asks "what would happen if we added two new categories to the existing 3-conclusion scale?" Because both scales included the Exclusion and Identification conclusions, we provided explicit definitions of these terms as illustrated in Table 1. However, the second comparison conceptually asks "if we switched to a strength-of-support conclusion scale, how would the response distribution change?" In this case, we asked participants to use the definitions that they had refined during casework, and we did not provide definitions for the traditional terms. We will show that this difference across conditions did not meaningfully change how they interpreted the traditional terms (discussed in Figure 8 as part of the results section).

2.4.2 | Footwear

The instructions and procedure for the Footwear examiners were similar to those for Fingerprint examiners but included the instructions shown in Table 5. The interface shown in Figure 3 allowed Footwear examiners to conduct examinations. The images could be rotated, and the known image could be colorized and dragged over the questioned impression. Either image could be toggled on and off to aid in the comparison process. No marks were allowed in this interface. The design included a 30-minute timer and a pause button.

2.4.3 | Toolmarks

The instructions and procedure for the Toolmark examiners were similar to those for Fingerprint examiners but included the instructions shown in Table 6. The interface included a split screen

TABLE 5 Instructions given to footwear examiners.

You will be completing 60 footwear comparisons using an online interface we have developed for this purpose. Please make the following assumptions about the known shoe/boot:
1. The shoe was recovered almost immediately after the crime was committed, and so, you should assume that there was *no opportunity* for wear or alteration to occur on the shoe.
2. Each trial consists of a questioned image on left, a gel test impression of the suspect's shoe in the middle, and a photograph of the suspect's shoe on the right.
3. You may observe differences due to variable pressure between the two impressions. This results from the fact that the technician who made the test impressions did not know what pressure the criminal used when placing the mark. There may also be slight distortion in the photographs from the vice used to hold the shoe for photography.You will be using different scales on different trials, which will allow us to compare the two scales. We would like you to use one of the following two scales when making your conclusions, and we will tell you which scale you will use at the start of each trial.
The definitions for both conclusion scales are below. You are welcome to print this page if you would like these definitions to be available during your comparisons.

controlled by a slider. The questioned image (on the left in Figure 4) could be dragged around, as could the known impressions on the right in Figure 4. The two known impressions were clearly separated with a red line to differentiate each. The images could be rotated and zoomed. No participant markings were allowed in this interface. The design included a 30-min timer and a pause button.

3 | RESULTS

All images, data, and analysis code are available at the OSF repository: https://osf.io/xmwqg/?view_only=f1b996eee77d45d0907ecebdaa27437d. Table 7, Table 8, Table 11, and Table 12 provide the response distributions for the various conditions, which are discussed below. In principle, the each row in Table 7, Table 8, Table 11, and Table 12 will sum to a multiple of the number of participants multiplied by 15, which was the number of trials per participant in that row. However, a rare data-saving problem of unknown origin (likely due to intermittent network problems) resulted in 8 fingerprint examiners with 59 trials instead of 60 (four in each participant group), 3 footwear examiners with 59 trials instead of 60, and no toolmark examiners with missing data. No participant lost more than one trial. In addition, one toolmark-mated pair was incorrectly identified as nonmated when images were assigned to participants. This error was corrected during the analysis and resulted in one additional mated pair and one fewer nonmated pair trial assigned to each participant. While these issues are regrettable, the modeling section (described below) is almost completely unaffected by this unequal distribution of responses across mated and nonmated trials because we are comparing across the two scales, which were both affected by these issues, and because the modeling relies on frequencies not raw trial counts.

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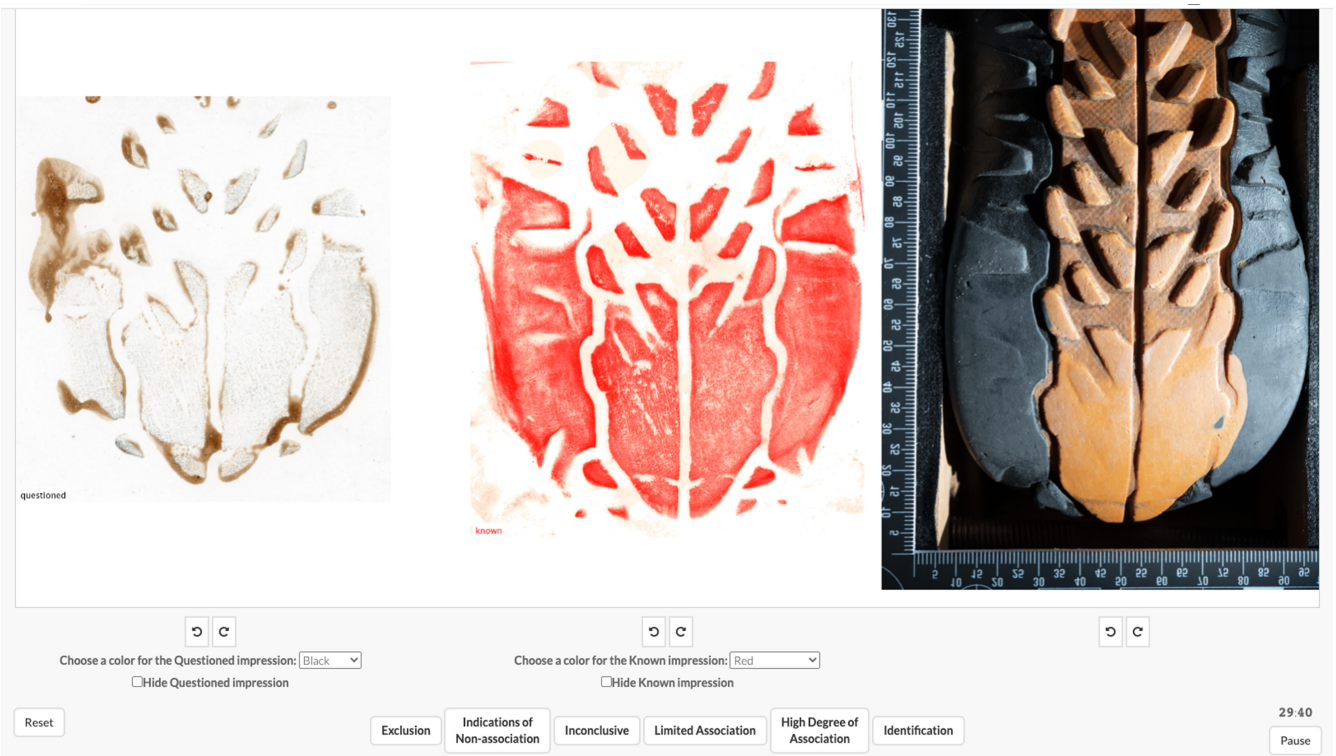


FIGURE 3 Interface used by footwear examiners to conduct casework-like comparisons [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 6 Instructions given to toolmark examiners

You will be completing 60 tool mark comparisons using an online interface we have developed for this purpose. Please make the following assumptions about the known tool:

- The tool was recovered almost immediately after the crime was committed, and so you should assume that there was *no opportunity* for wear or alteration to occur on the tool.
- There will be two test impressions in the comparisons, one at 20° and one at 35°. From the crime scene you are able to ascertain that the tool was used at an angle that falls within this range.
- You may observe differences due to variable pressure between the two impressions. This results from the fact that the technician who made the test impressions did not know what pressure the criminal used when using the tool.
- The tools included in the dataset are 1/4 "screwdrivers and 1/4" chisels. All are impressed on heavy-duty aluminum foil. You should make no assumptions about the questioned impression, other than it is either a screwdriver or a chisel, nor should you assume that each trial contains *only* screwdrivers or chisels. Some trials may contain a questioned mark from a screwdriver, and test impressions from a chisel, for example. However, both test impressions were definitely made by the same tool, just at different angles.

3.1 | Response distributions

3.1.1 | Fingerprint examiners

Table 7 illustrates the response distribution for participants who compared the traditional scale with the strength-of-support scale. Table 8 illustrates the response distribution for participants who compared the traditional scale with the expanded traditional scale. There were six erroneous identification responses across the two groups (5 erroneous identification outcomes and 1 erroneous extremely strong support for common source outcome). Four pairs had one erroneous identification outcome, while one pair had two erroneous identification outcomes. In each case, the ground truth was verified against the original scans to verify the nonmated status of each of the five nonmated pairs that produced erroneous identification outcomes. Combining over both participant groups and scale types, there were 14 erroneous Support for Common Source outcomes (8 from the strength-of-support scale and 6 from the expanded traditional scale). There were 86 erroneous exclusion or erroneous extremely strong support for different source outcomes (26 and 30 erroneous exclusions from the traditional scale from the two participant groups, 22 erroneous exclusions from the expanded traditional scale, and 8 erroneous extremely strong support for different source outcomes from the strength-of-support scale).

Consistent with Carter, Vogelsang [8], the number of correct identification outcomes dropped when the scale was expanded. This was more pronounced in the Traditional/strength-of-support comparison (245–174 in Table 7) and somewhat pronounced in the Traditional/Expanded Traditional comparison (231–199 in Table 8). These results are again consistent with the finding that examiners redefine the definition of the term Identification when the scale is expanded (see Table 8) or are less likely to use Extremely Strong Support for Common Source than the term Identification (see Table 7).

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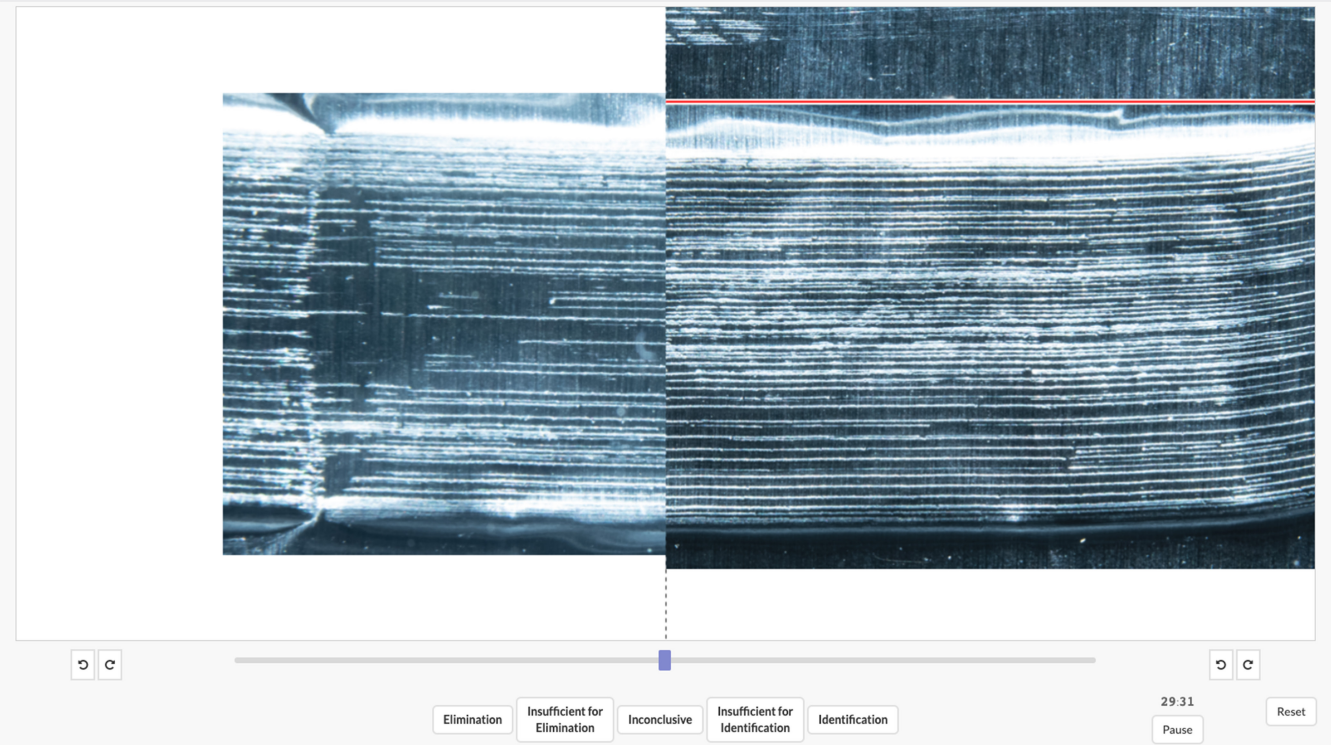


FIGURE 4 Interface used by toolmark examiners to conduct casework-like comparisons [Color figure can be viewed at wileyonlinelibrary.com]

The number of Inconclusive responses to mated pairs also dropped as the traditional scale was expanded. These dropped from 220 to 131 in Table 8, and from 238 to 176 in Table 7. These results demonstrate that the Support for Common Source response to mated pairs is a mixture of what would have been Inconclusive and Identification responses in the Traditional scale.

The number of Exclusion responses to nonmated pairs also dropped for the expanded scales. These responses dropped from 272 to 225 in Tables 8 and 258 to 192 in Table 7. This suggests that examiners become risk-averse for the expanded scales on the exclusion side.

The numbers in parantheses in Table 7 and Table 8 correspond to those trials where the participant rated the latent impression as either of value for exclusion or of value for identification. As can be seen by inspection, the counts change fairly dramatically for the Inconclusive conclusions, but conditioning on only impressions deemed "of value" does not markedly affect the other conclusion counts. It is of some interest that examiners made some correct Support for Common Source decisions even after labeling the latent print of no value. For example, in Table 7, examiners made 5 more correct decisions (114–109) and two more erroneous decision [8–6]. They also made two more correct identifications in the traditional scale (245–243). In Table 8, examiners made 4 more correct identifications in the traditional scale, 13 more correct Support for Common Source decisions (93–80), and one more erroneous Support for Common Source decision [6–5]. While there may be operational reasons for thresholding latent prints with the "No Value" decision, clearly, there are cases where a marginal latent print turns out to be useful upon inspection of the comparison print.

Table 9 contains the response proportions for the traditional/strength-of-support comparison group of examiners, which is created by dividing the numbers in Table 7 by the row sum. The numbers in parantheses correspond to the proportions for data where "no value" trials are excluded. Table 10 contains the response proportions for the traditional/strength-of-support comparison group of examiners, which is created by dividing the numbers in Table 8 by the row sum. These proportions can be used to determine derived measures such as specificity and sensitivity. Note, however, that these measures are nonlinear, and the signal detection theory criterion estimates shown in Figures 5 and 6 are more useful when comparing across scales. Equal distances on the evidence axes in Figures 5 and 6 are meaningful comparisons because the criterion estimates on the decision axis tend to be linear with psychological constructs such as ability or risk aversion.

One consequence of moving to an expanded scale is that suboptimal outcomes may occur with the Support for Common Source conclusion and nonmated pairs. In prior work [8], we found 17 erroneous support for common source conclusions (i.e., saying support for common source to a nonmated pair) out of a total of 408 nonmated comparisons, for an error rate of 4.17%. The current study used the same images but different examiners, and we found 6 erroneous support for common source conclusions out of 479 trials for the expanded traditional scale and 8 out of 509 trials for the strength-of-support scale. These error rates are 1.25% and 1.57%, respectively. It may be that examiners are becoming more familiar

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TABLE 7 Response distribution for fingerprint participants in the experimental group that compared the traditional scale with a pure strength-of-support scale. Numbers in parantheses correspond to the count of outcomes limited to trials where the latent impressions were not judged of no value

		Traditional scale		
Ground truth	**Exclusion**		**Inconclusive**	**Identification**
Nonmated	258 (251)	NA	250 (163)	1 (1)
Mated	26 (22)	NA	238 (191)	245 (243)

	Strength-of-support scale			
Ground truth	Extremely strong support for different sources	Support for different sources	Inconclusive	Support for common source	Extremely strong support for common source
Nonmated	192 (190)	125 (116)	183 (110)	8 (6)	1 (1)
Mated	8 (8)	37 (34)	176 (125)	114 (109)	174 (174)

with expanded conclusion scales and are learning to use these conclusions more appropriately.

3.1.2 | Footwear examiners

Table 11 provides the response distributions for footwear examiners. There were five total erroneous identification outcomes distributed across five different nonmated sets. In each case, the ground truth was verified by accessing the raw images collected from the scanner or photography rig that contained the shoe pair number, and in each case, these nonmated pairs were verified as coming from different shoes. However, as previously noted, many of these shoes are not only of the same make and model, but also were worn by the same individual and retired with similar wear because of the nature of the running activity. This difficulty may not be fully representative of casework as a result, but the results still allow for comparisons across scales.

Both the Traditional and strength-of-support scales have six categories, and the question of interest is whether the response distribution changes as the scale changes. We might find, for example, that examiners are reluctant to use a particular statement such as Extremely Strong Support for Common Source. However, Table 11 demonstrates that there were no large differences between the two conclusion scales, with perhaps a slight drop between High Degree of Association and Strong Support for Common Source. Thus, these scales seem to be treated more or less equivalently by participants.

One reason for this lack of strong shift in response distributions across the two scales may be the nature of the existing Footwear scale. Note that the existing footwear articulation language in Table 3 is arguably more similar to the strength-of-support conclusion scale than the Fingerprint or Toolmark traditional scales are to the strength-of-support scales because the existing Footwear scale includes concepts of degree of association rather than statements about source. The footwear scale still includes Identification, but the definition of Identification is phrased in terms of likelihood of observations.

3.1.3 | Toolmark examiners

Table 12 provides the response distributions for the toolmark examiners. There were four erroneous identification outcomes, which were distributed across four different nonmated pairs. In each case, the ground truth was verified against the original scans to verify the nonmated status of each of the four pairs. It is difficult to establish the task difficulty of these comparisons relative to casework, although the fact that the toolmarks were created by tools of the same make and model does make this a particularly challenging task. However, the response distributions do allow for comparisons across scales, which is the intent of the study.

Both the Traditional scale and the strength-of-support scale have 5 statements, and the question of interest is whether the response distributions are similar across the two scales. That is, do examiners treat the two scales in the same way? We find that Extremely Strong

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TABLE 8 Response distribution for fingerprint participants in the experimental group that compared the traditional response scale with the expanded traditional scale. Numbers in parentheses correspond to the count of outcomes limited to trials where the latent impressions were not judged of no value

Traditional scale					
Ground truth	**Exclusion**		**Inconclusive**		**Identification**
Nonmated	272 (261)	*NA*	207 (129)	*NA*	2 (2)
Mated	30 (26)	*NA*	220 (157)	*NA*	231 (227)
Expanded traditional scale					
Ground truth	Exclusion	Support for different sources	Inconclusive	Support for common source	Identification
Nonmated	225 (213)	92 (81)	154 (86)	6 (5)	2 (2)
Mated	22 (18)	33 (31)	131 (89)	93 (80)	199 (199)

Support for Common Source demonstrates a degree of risk aversion because participants used this response category for mated pairs much less often than the Identification response (136 vs. 177). Most of these responses that might have been Identification responses in the Traditional scale appear to have been moved to the Support for Common Source because the outcomes grew from 47 for Insufficient for Identification to 84 in the Support for Common Source.

Evidence for risk aversion in the exclusion outcomes is evident for the strength-of-support scale because the number of correct exclusions drops from 84 in the Traditional scale to 45 in the strength-of-support scale.

3.2 | Estimating decision criterion

The response proportions presented in Table 9 through Table 12 can be summarized using extensions to Signal Detection Theory. As in Carter, Vogelsang [8], we fit the response distributions with a model that assumes that the result of each comparison produces a unidimensional value on an internal evidence axis, which is then mapped to a categorical statement using a set of decision criteria. The distribution of nonmated and mated pairs along this evidence axis is summarized using Gaussian distributions, and separate decision criteria are fit to each scale. Further details are found in Carter, Vogelsang [8], but for the present work, we fit the combined data across all subjects using the brms library [13] in R [14].

The goal of signal detection theory is to separate ability (as measured by d') from response aversion/bias (as measured by the decision criteria). Although it is possible for sensitivity (d') to differ across scales, prior work found no evidence for this, and thus, we first established a common d' and standard deviation value for the mated pair distribution and then fit separate decision criteria for each scale. The reason we fit these models is that they convert nonlinear measures such as sensitivity and specificity to values on an axis that is likely to be linear (the evidence axis of signal detection theory). Equal distances on the evidence axis correspond to equal shifts in the decision criteria, regardless of the absolute values of sensitivity and specificity.

The results of the modeling for each participant group are provided below. We fit the combined data for each group rather than individual participants because we are interested in how the field as a whole would respond if the conclusion scale were changed. In our earlier work with a very similar design, we fit individual participants in addition to group data and found similar results across the two types of fits [8].

3.2.1 | Fingerprint examiners

The sensitivity (d') value for fingerprint examiners across the two participant groups was 2.39, with a standard deviation for the mated pair distribution of 1.48. This difficulty level is consistent with other error rate studies (Ulery, Hicklin [11]; see Mannering, Vogelsang [10]) and thus the task difficulty appears similar to that of casework. Figure 5 illustrates the results of this modeling, and shows the location of different decision criteria for the two scales. The color bands represent 95% confidence intervals around the decision criterion estimates. As was suggested by the response distributions in Table 8, the decision criteria for Identification in the Expanded scale is shifted to the right of the Identification decision criteria for the Traditional scale. This is consistent with prior results [8] and provides evidence that examiners become more risk-averse with expanded scales. A similar result is found with Exclusion, where examiners are less likely to use this response category in the expanded scale, thus pushing the Exclusion decision criteria from the Expanded scale to the left.

Examiners become even more risk-averse when asked to use the Extremely Strong Support for Common Source conclusion, as shown in Figure 6. They are also less likely to use the Extremely Strong Support for Different Sources conclusion than the Exclusion conclusion. In each case, examiners become more risk-averse with the expanded scale. The Inconclusive inconclusive area in the traditional scale also shrinks when the scale is expanded. These results demonstrate that examiners reserve conclusion statements that include Extremely Strong Support for only those conclusions with the highest amount of support.

Although we did not compare the Expanded Traditional and strength-of-support scales directly, we can do a virtual comparison across participant groups because our model fits rely on a common d' and mated distribution standard deviation across the two participant groups. Figure 7 illustrates the decision criteria for the two five-item scales and demonstrates that examiners are more risk-averse when using the strength-of-support scale than the expanded traditional scale. This is again consistent with the above result that suggests that

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TABLE 9 Response proportions for fingerprint participants in the experimental group that compared the traditional scale to a pure strength-of-support scale. Numbers in parantheses correspond to the proportion of outcomes limited to trials where the latent impressions were not judged of no value

	Traditional scale			
Ground truth	Exclusion		Inconclusive	Identification
Nonmated	0.507 (0.605)	NA	0.491 (0.393)	NA	0.002 (0.002)
Mated	0.051 (0.048)	NA	0.468 (0.419)	NA	0.481 (0533)
	Strength-of-support scale			
Ground truth	Extremely strong support for different sources	Support for different sources	Inconclusive	Support for common source	Extremely strong support for common source
Nonmated	0.377 (0.449)	0.246 (0.274)	0.360 (0.260)	0.016 (0.014)	0.002 (0.002)
Mated	0.016 (0.018)	0.073 (0.076)	0.346 (0.278)	0.224 (0.242)	0.342 (0.387)

any conclusion statement that contains Extremely Strong Support is reserved for comparisons with the highest amount of support.

Finally, note that there were subtle differences in the instructions given the two participant groups with respect to the use of the Identification, Inconclusive, and Exclusion terms. This was done deliberately because the data from the two groups are being used to address slightly different scientific questions. However, we see that the two groups performed very similarly with respect to the placement of their decision criteria, as shown in Figure 8. The Identification criteria are almost identical, and the Exclusion criteria are also quite similar. Thus, we feel that the subtle differences in instructions still allow for comparisons across the two groups as in Figure 7.

3.2.2 | Footwear examiners

The fitted values of d' is 2.14, and the standard deviation for the mated pairs was 1.13. The response distributions across the two scales shown in Table 11 demonstrated that there were no large shifts in responses across the two scales. Consistent with this result, Figure 9 demonstrates that the fitted decision criteria across the two scales were very similar, with perhaps a slight difference between the High Degree of Association and Strong Support for Common Source decision criteria. Thus, it appears that footwear examiners treat these two scales in a very similar manner.

3.2.3 | Toolmark examiners

The fitted values of d' is 2.49, and the standard deviation for the mated pairs was 1.77. The response distributions in Table 12 illustrated that toolmark examiners grew more risk-averse when using the strength-of-support scale. As shown in Figure 10, the fitted decision criteria for the Extremely Strong Support for Common Source is to the right of the Identification decision criterion, demonstrating increased risk aversion for the strength-of-support scale. This is also true for the Exclusion/Extremely Strong Support for Different Sources comparison. It appears that toolmark examiners become much more risk-averse when using the strength-of-support scale. This result is consistent with that observed with Fingerprint examiners, although Footwear examiners did not show evidence of such a shift.

4 | DISCUSSION

The present work provides four clear conclusions:

1. Fingerprint examiners became more risk-averse when moving from the traditional scale to the strength-of-support scale. Surprisingly, they show a strong reticence to use the Extremely Strong Support for Common Source conclusion relative to their use of Identification in the traditional scale. We view this as surprising, because in a companion article (T. Busey and M.

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TABLE 10 Response proportions for fingerprint participants in the experimental group that compared the traditional response scale to the expanded traditional scale. Numbers in parantheses correspond to the proportion of outcomes limited to trials where the latent impressions were not judged of no value

Traditional scale					
Ground truth	Exclusion		Inconclusive		Identification
Nonmated	0.567 (0.666)	NA	0.429 (0.329)	NA	0.004 (0.005)
Mated	0.063 (0.063)	NA	0.455 (0.383)	NA	0.482 (0.554)
Expanded traditional scale					
Ground truth	Exclusion	Support for different sources	Inconclusive	Support for common source	Identification
Nonmated	0.470 (0.550)	0.192 (0.209)	0.322 (0.222)	0.013 (0.013)	0.004 (0.004)
Mated	0.046 (0.043)	0.069 (0.074)	0.274 (0.213)	0.195 (0.192)	0.416 (0.477)

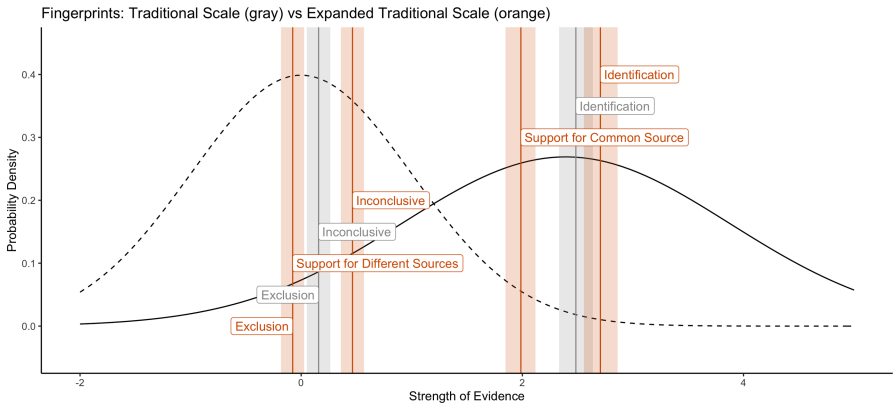


FIGURE 5 Estimates of the decision criteria for the comparison for Fingerprint examiners between the Traditional 3-item scale (Exclusion/Inconclusive/Identification) with the Expanded Traditional 5-item scale that included the Support For Different Sources and Support for Common Source categories. Color bands represent 95% confidence intervals. Note that the Identification criterion shifts to the right for the expanded scale (red) relative to the traditional scale (gray), indicating that examiners use this category less often than when they have only three categories to choose from. Examiners are also more risk-averse when making Exclusion conclusions when using the expanded scale [Color figure can be viewed at wileyonlinelibrary.com]

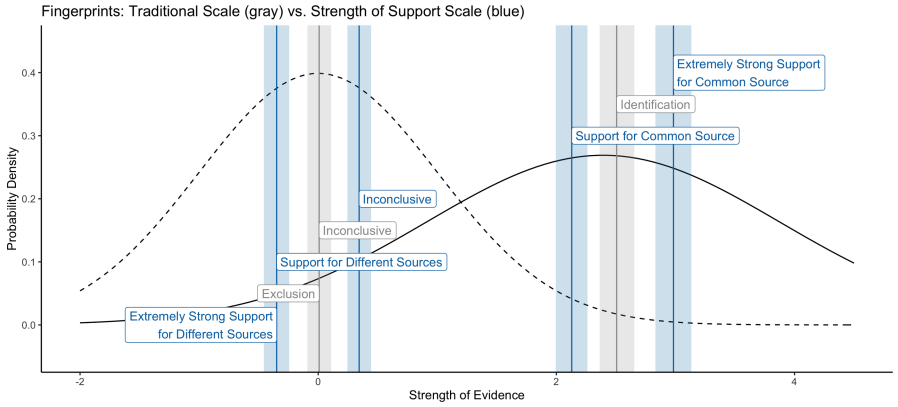


FIGURE 6 Estimates of the decision criteria for the comparison between the Traditional three-item scale (Exclusion/Inconclusive/Identification) with the strength-of-support three-item scale for Fingerprint examiners. Examiners become more risk-averse when using the expanded strength-of-support scale (see text for details) [Color figure can be viewed at wileyonlinelibrary.com]

Klutzke, personal communication, https://buseylab.sitehost.iu.edu/homepage/publications/PDFS/BuseyAndKlutzkeCalibratingEvidenceMaster.pdf, February 10, 2022) we found that examiners viewed Extremely Strong Support for Common Source as implying *less* evidence than Identification for the proposition of common source when comparing the two on a visual scale. This disconnect perhaps reduces the utility of the Extremely Strong Support for Common Source conclusion as a policy recommendation because examiners might use it less often, yet think it means something less than it does. Members of the general public, however, interpret it as equivalent to Identification as discussed in the companion article referenced above, further reducing the utility of this term as a conclusion scale statement.

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TABLE 11 Response distribution for footwear examiners. Numbers in parantheses correspond proportion of outcomes in that cell, where the rows sum to 1.0 (subject to rounding error)

Definitive conclusions (traditional) scale						
Ground truth	Exclusion	Indications of non-association	Inconclusive	Limited association	High degree of association	Identification
Nonmated	260 (0.540)	126 (0.264)	11 (0.023)	65 (0.136)	15 (0.031)	3 (0.006)
Mated	25 (0.052)	23 (0.048)	17 (0.035)	117 (0.244)	146 (0.305)	151 (0.315)
Strength-of-support scale						
Ground truth	Extremely strong support for different sources	Support for different sources	Indeterminate with respect to source	Support for common source	Strong support for common source	Extremely strong support for common source
Nonmated	258 (0.535)	123 (0.256)	33 (0.069)	51 (0.104)	15 (0.031)	2 (0.004)
Mated	18 (0.038)	29 (0.060)	23 (0.048)	147 (0.306)	103 (0.215)	160 (0.333)

TABLE 12 Response distribution for toolmark examiners. Numbers in parantheses correspond proportion of outcomes in that cell, where the rows sum to 1.0 (subject to rounding error)

Traditional scale					
Ground truth	Elimination	Insufficient for elimination	Inconclusive	Insufficient for identification	Identification
Nonmated	84 (0.289)	68 (0.234)	113 (0.388)	24 (0.082)	2 (0.007)
Mated	8 (0.026)	20 (0.065)	57 (0.184)	47 (0.152)	177 (0.573)
Strength-of-support scale					
Ground truth	Extremely strong support for different sources	Support for different sources	Inconclusive	Support for common source	Extremely strong support for common source
Nonmated	45 (0.156)	119 (0.412)	110 (0.381)	13 (0.045)	2 (0.007)
Mated	11 (0.035)	19 (0.061)	61 (0.196)	84 (0.270)	136 (0.437)

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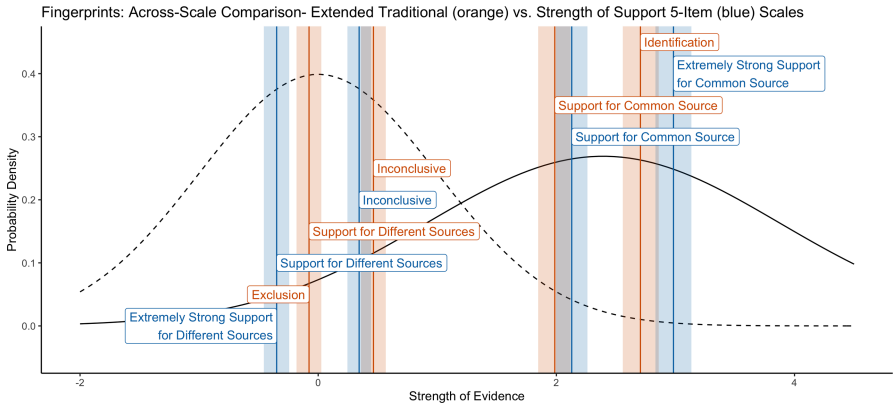


FIGURE 7 Across-scale comparison between the two five-item scales for Fingerprint examiners. This comparison combines the data from both participant groups to estimate how each scale would be used if adopted for casework. Color bands represent 95% confidence intervals. The strength-of-support scale tends to make examiners more risk-averse because the Extremely Strong Support for Common Source decision criterion is to the left of the Identification decision criteria. This results from the fact that examiners use Extremely Strong Support for Common Source less often than Identification [Color figure can be viewed at wileyonlinelibrary.com]

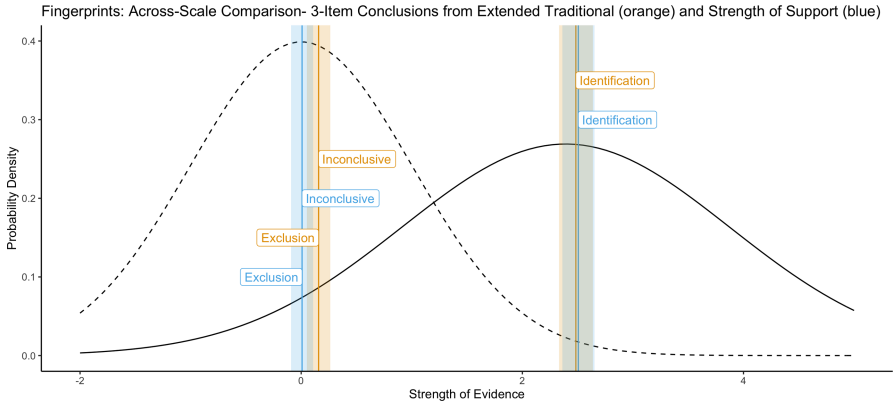


FIGURE 8 Comparison of estimates for decision criteria for the two three-item scales for Fingerprint examiners. Color bands represent 95% confidence intervals. The two sets of participants had slightly different instructions for the three-item scales (one provided explicit definitions, while the other asked them to use whatever criteria they would apply to a three-item scale in casework). This graph illustrates that the estimates for the Identification criteria are almost identical, while there is slight variation in the Exclusion criteria [Color figure can be viewed at wileyonlinelibrary.com]

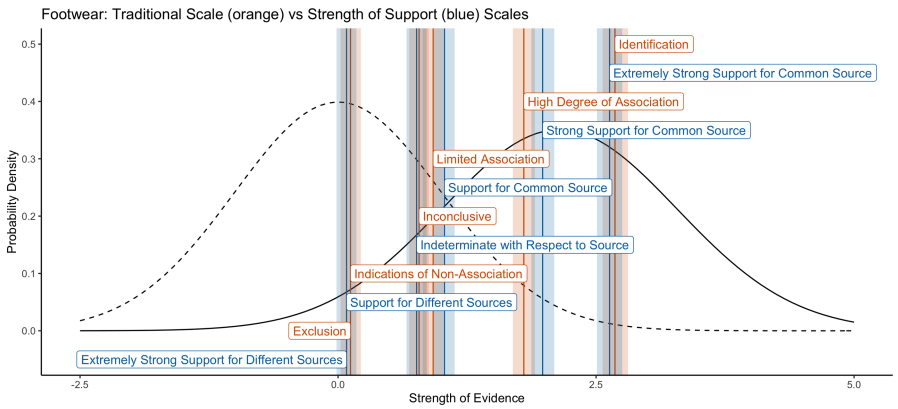


FIGURE 9 Estimates of the decision criteria for the comparison for the Traditional Scale and the strength-of-support scale for Footwear examiners. Color bands represent 95% confidence intervals. The two scales appear to be used similarly, with perhaps a slight difference between High Degree of Association and Strong Support for Common Source [Color figure can be viewed at wileyonlinelibrary.com]

One possibility that we did not test is whether Strong Support for Common Source (without the term "Extremely") might be a more appropriate endpoint to a strength-of-support scale. Examiners may show less risk aversion to using this phrase, and this phrase may be more justified given the error rate studies that show an erroneous identification rate of 0.1% [11].

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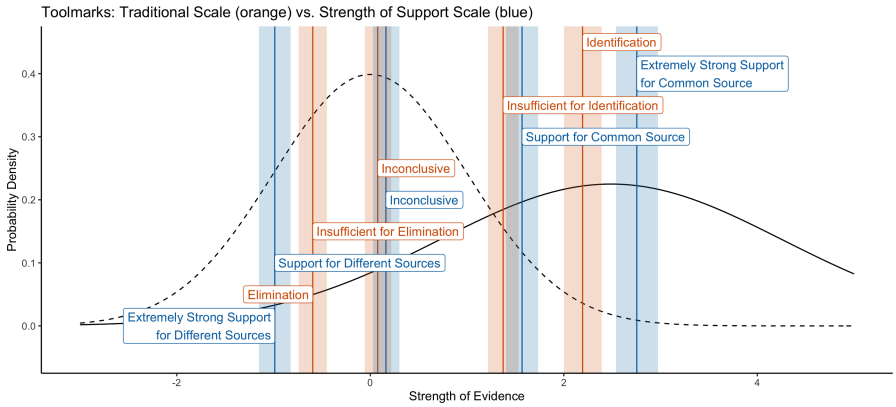


FIGURE 10 Estimates of the decision criteria for the Traditional Scale (orange) and the strength-of-support scale (blue) for Toolmark examiners. Color bands represent 95% confidence intervals. Examiners become more risk-averse when using the expanded strength-of-support scale (see text for details) [Color figure can be viewed at wileyonlinelibrary.com]

2. In footwear comparisons, the examiners did not exhibit a large shift in responses from the traditional to strength-of-support scale. However, given the nature of footwear impressions, the use of the Extremely Strong Support for Common Source may be inappropriate depending on what magnitude of LRs can be generated given the discriminability of the footwear characteristics, although this deserves further consideration as future error rate studies are completed. Thompson and Newman [15] found that prior beliefs about a discipline affect evidence interpretation by mock jurors, and so, even if members of the general public interpret the highest categories on different scales as equivalent, they will probably contextualize this result given their understanding of the individual discipline.
3. Toolmark examiners exhibited strong risk aversion when using the strength of support conclusion scale, similar to that observed with fingerprint examiners (see Figure 10). As with fingerprint comparisons, we suggest that perhaps Extremely Strong Support for Common Source is too strong relative to the strength of the evidence in a discipline, and that the discipline might consider Strong Support for Common Source as the highest category of conclusion statements.
4. In fingerprint comparisons, participants redefined the term Identification when Support for Common Source was included in the conclusion scale, relative to the traditional scale. This is a direct replication of the Carter, Vogelsang [8] result. The Support for Common Source category absorbed some of the weaker Identification conclusions from the traditional scale, as well as some of the stronger Inconclusive conclusions from the traditional scale. We view both of these as positive outcomes because perhaps some of the weaker identifications may have been borderline and arguably at the boundary of sufficiency for Identification, and some of the stronger Inconclusive conclusions probably merited an investigative lead.

In general, we view strength-of-support scales as an improvement over scales with traditional language, for reasons discussed in [1]. However, the consumer must be taught to interpret the conclusion scale properly, which should include saying what could have been concluded but was not. For the fingerprint discipline, there are also other operational considerations when expanding a scale. For example, examiners may have to work longer to reach an Identification (or Extremely Strong Support for Common Source) conclusion than a Support for Common Sources conclusion and may decide to terminate the examination process earlier if they can make a less definitive conclusion when using an expanded scale. This interacts with the laboratory's current backlog and how consumers of that agency use less definitive conclusions, and therefore, our data do not bear directly on what might happen operationally should an expanded scale be introduced. We suggest that each laboratory conducts its own studies to determine the possible effect of expanded scales given their own policies and constraints. For example, forensic laboratories with large backlogs may benefit from making relatively rapid investigative lead conclusions if full Identification conclusions would take extensive time and effort.

strength-of-support scales in two of the three disciplines resulted in a shift of examiner behavior toward becoming more risk-averse because participants used the Extremely Strong Support for Common Source conclusion less often than Identification. Thus, while strength-of-support language may focus on the evidence rather than the examiner (i.e., "the evidence supports" as opposed to "I identified"), the words Extremely Strong Support may not be justified by the error rate studies in a given discipline, and the examiners may have understood this by using this term infrequently and only for the strongest cases. Consideration should be given to how the consumers might interpret various statements [16].

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